



หัวข้อวิทยานิพนธ์สำหรับหลักสูตรวิศวกรรมศาสตรดุษฎีบัณฑิต สาขาวิชาวิศวกรรมคอมพิวเตอร์
ประจำปีการศึกษา 2569

- ☒ 4236 (สำหรับผู้สำเร็จปริญญาโทที่ต้องการทำวิทยานิพนธ์อย่างเดียว)
- ☒ 4237 (สำหรับผู้สำเร็จปริญญาตรีที่ต้องการทำวิทยานิพนธ์อย่างเดียว)
- ☒ 4238 (สำหรับผู้สำเร็จปริญญาโทที่ต้องการเรียนรายวิชาและทำวิทยานิพนธ์)
- ☒ 4239 (สำหรับผู้สำเร็จปริญญาตรีที่ต้องการเรียนรายวิชาและทำวิทยานิพนธ์)

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ช่องทางการติดต่ออื่นๆ	https://cragkhit.github.io https://cuselab.github.io/

ชื่อหัวข้อ (ไทย)	การออกแบบชุดทดสอบซอฟต์แวร์อัตโนมัติจาก Business Process Model and Notation (BPMN)
ชื่อหัวข้อ (อังกฤษ)	Automated Software Test Generation using Business Process Model and Notation (BPMN)
เนื้อหาโดยสังเขป	Business Process Model and Notation (BPMN) bridges the communication gap between business stakeholders and technical teams by formally defining system behaviors and workflows. While model-based testing leverages BPMN for quality assurance, traditional techniques struggle to bridge the abstraction gap between high-level process flows and concrete, executable test scripts. This PhD research proposes a hybrid framework for automatically generating executable system and integration test suites, including dynamic test data, from BPMN business process specifications. The project will (1) formulate a semantic enrichment model that maps BPMN process constructs, gateways, and data objects to executable interface definitions (e.g., REST APIs, microservice interactions, and UI workflows), (2) develop an automated path-selection and test-data generation strategy that combines formal model execution with AI-driven synthesis to cover edge cases, complex decision logic, and exception flows, (3) synthesize end-to-end executable test scripts (e.g., PyTest, Playwright, or Postman) and evaluate the approach on real-world enterprise business processes, assessing path coverage, test execution reliability, and fault-detection effectiveness.

ชื่อหัวข้อ (ไทย)	AI Coding Agent ที่คำนึงถึงคุณภาพของซอฟต์แวร์
ชื่อหัวข้อ (อังกฤษ)	Software Quality-Aware AI Coding Agents
เนื้อหาโดยสังเขป	The rapid adoption of AI coding agents has accelerated software development, yet code generated by these models frequently introduces technical debt, code smells, and maintainability issues. Current AI agents prioritize functional correctness over structural quality, creating long-term

	maintenance burdens. This PhD research aims to develop software quality-aware AI coding agents capable of real-time quality evaluation and autonomous code refactoring during generation. The project will (1) formulate a multi-dimensional software quality model tailored for AI-generated code, focusing on maintainability metrics such as technical debt, cognitive complexity, and coding conventions, (2) design a dynamic, in-the-loop quality assessment mechanism integrating static analysis and LLM self-reflection during token and block-level synthesis, (3) establish closed-loop decision-making frameworks (e.g., multi-agent critic-developer loops and dynamic self-repair strategies) to trigger real-time refactoring when quality degradation is detected. The proposed approach will be implemented as an open agentic framework and benchmarked against state-of-the-art coding agents to evaluate its impact on code maintainability, execution correctness, and developer productivity.
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ชื่อหัวข้อ (ไทย)	ชุดเครื่องมือในการทำและประเมินตนเองตามมาตรฐาน ISO/IEC 29110 แบบอัตโนมัติด้วยโมเดลภาษาขนาดใหญ่
ชื่อหัวข้อ (อังกฤษ)	Automated Tool Suite for the Implementation and Self-Auditing of ISO/IEC 29110 Using Large Language Models
เนื้อหาโดยสังเขป	ISO/IEC 29110 is an international standard suite designed to assist Very Small Entities (VSEs), organizations or teams with fewer than 25 people, in adopting standardized systems and software life cycle processes. While widely adopted in regions such as Thailand, the standard imposes significant documentation overhead and manual audit efforts that strain VSE resources. Leveraging Large Language Models (LLMs), this PhD research proposes a domain-specific, hallucination-resistant framework for automating document generation, process adherence, and self-auditing within ISO/IEC 29110 workflows. The project will (1) formulate a semantic mapping model that translates ISO/IEC 29110 process profiles (e.g., Project Management and Software Implementation) into structured, machine-interpretable prompts and constraints, (2) design an intelligent multi-agent framework utilizing Retrieval-Augmented Generation (RAG) and deterministic verification guardrails to automatically generate compliant project artifacts and detect process non-conformances, (3) develop an integrated tool suite and evaluate its performance across software VSEs, measuring audit precision, documentation time reduction, and artifact consistency against human expert assessments.

ชื่อหัวข้อ (ไทย)	วิศวกรรมซอฟต์แวร์เพื่อโปรแกรมควอนตัม
ชื่อหัวข้อ (อังกฤษ)	Software Engineering for Quantum Programs (SE4Q)
เนื้อหาโดยสังเขป	Quantum computing is rapidly transitioning from theoretical frameworks to practical application, giving rise to complex quantum software systems. However, existing software engineering practices designed for classical systems are often ill-suited for quantum paradigms due to fundamental quantum

	phenomena such as superposition, entanglement, and non-determinism. Consequently, quantum developers face significant hurdles in testing, debugging, and maintaining reliable quantum code. This PhD research focuses on establishing rigorous software engineering principles, methodologies, and automated tools specifically designed for quantum programs. The work will first analyze common fault patterns, quality attributes, and maintainability challenges in existing quantum software development. Next, it will design novel techniques for quantum program verification, automated test case generation, and quality assessment. Finally, the proposed methodologies will be implemented as a specialized tool suite and evaluated against benchmark quantum algorithms and real-world quantum software platforms.
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ชื่อหัวข้อ (ไทย)	ควอนตัมสำหรับวิศวกรรมซอฟต์แวร์
ชื่อหัวข้อ (อังกฤษ)	Quantum Computing for Software Engineering (Q4SE)
เนื้อหาโดยสังเขป	Software engineering involves numerous computationally intensive tasks, such as automated test suite optimization, program synthesis, bug localization, and software architecture refactoring, that reduce to NP-hard optimization or search problems. As modern software systems grow in scale and complexity, classical heuristics and search-based software engineering (SBSE) techniques increasingly face performance bottlenecks. Quantum computing, leveraging algorithms such as Quantum Annealing and the Quantum Approximate Optimization Algorithm (QAOA), offers a promising paradigm shift for solving these intractable software engineering challenges with potential computational speedups. This PhD research investigates the application of quantum computing and quantum-inspired algorithms to solve complex software engineering problems. The work will first map specific high-complexity software engineering tasks into quantum-compatible mathematical frameworks, such as Quadratic Unconstrained Binary Optimization (QUBO). It will then develop novel hybrid classical-quantum algorithms tailored for tasks like automated test case prioritization, dependency resolution, and system refactoring. Finally, the proposed quantum approaches will be benchmarked against state-of-the-art classical techniques using real-world software engineering datasets to evaluate their efficiency, solution quality, and scalability.